

7.1 Linear unbiased estimation

To call an estimator the best, we have to say what it is compared with and by what criterion. The comparison can't be made against every estimator. The constant estimator $\tilde{\beta}_1 \equiv 0.5$ of a slope has variance zero, and it is perfect when the true slope happens to be 0.5 and useless otherwise. No estimator has smaller variance, yet nobody would call it good. Some restriction on the class of competitors is unavoidable. The classical restriction has two parts. The estimator must be *unbiased*, which rules out estimators that buy precision by ignoring the data. And it must be *linear* in the response, which makes its variance computable from the first two moments of the errors alone. This section describes the class these two conditions carve out. The description is geometric, and the Gauss–Markov theorem of [Section 7.2](#) will follow from it in a few lines.

7.1.1 The setting

Throughout this chapter the response satisfies the linear model of [Chapter 5](#) (Definition 5.3.1),

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbf{E}(\boldsymbol{\varepsilon}) = \mathbf{0}, \quad \text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}, \quad (7.1.1)$$

with $\mathbf{X}$ a fixed $n \times p$ matrix of rank $r \leq p$, $\boldsymbol{\beta} \in \mathbb{R}^p$ unknown and $\sigma^2 > 0$ unknown. These are the **second-moment assumptions**. The errors have mean zero, a common variance, and are uncorrelated. Nothing is assumed about their distribution beyond that. From [Section 7.3](#) on we add normality, and we will say so each time. As in [Chapter 6](#), $\mathbf{M}$ denotes the orthogonal projection onto $\mathcal{C}(\mathbf{X})$, $\hat{\boldsymbol{\beta}}$ is any least squares estimate, and $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2$ is the residual sum of squares.

The quantities we want to estimate are linear functions $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$, for fixed $\boldsymbol{\lambda} \in \mathbb{R}^p$. This covers a single coefficient ($\boldsymbol{\lambda}$ a coordinate vector), a difference of two coefficients, and the mean response $\mathbf{x}_0^\top \boldsymbol{\beta}$ at a new point $\mathbf{x}_0$. Several such functions at once are collected as $\boldsymbol{\Lambda}^\top \boldsymbol{\beta}$, where $\boldsymbol{\Lambda}$ is a $p \times q$ matrix.

7.1.2 Linear and unbiased

Definition 7.1.1 — Linear unbiased and best linear unbiased estimators

Let $\boldsymbol{\lambda} \in \mathbb{R}^p$.

- (a) A **linear estimator** is a statistic of the form $c + \mathbf{a}^\top \mathbf{Y}$, with a constant $c \in \mathbb{R}$ and a constant vector $\mathbf{a} \in \mathbb{R}^n$ (constant means not depending on $\mathbf{Y}$ or on the parameters).
- (b) A linear estimator is a **linear unbiased estimator** (LUE) of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ if $\mathbf{E}(c + \mathbf{a}^\top \mathbf{Y}) = \boldsymbol{\lambda}^\top \boldsymbol{\beta}$ for every $\boldsymbol{\beta} \in \mathbb{R}^p$ and every $\sigma^2 > 0$.
- (c) An LUE $c + \mathbf{a}^\top \mathbf{Y}$ of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is a **best linear unbiased estimator** (BLUE) if $\text{Var}(c + \mathbf{a}^\top \mathbf{Y}) \leq \text{Var}(d + \mathbf{b}^\top \mathbf{Y})$ for every LUE $d + \mathbf{b}^\top \mathbf{Y}$ of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$, for every $\boldsymbol{\beta}$ and every $\sigma^2 > 0$.
- (d) For q functions $\boldsymbol{\Lambda}^\top \boldsymbol{\beta}$, a **linear estimator** is $\mathbf{c} + \mathbf{A}\mathbf{Y}$ with a constant $q \times n$ matrix $\mathbf{A}$

and $\mathbf{c} \in \mathbb{R}^q$. It is an LUE of $\mathbf{\Lambda}^\top \boldsymbol{\beta}$ if each coordinate is an LUE of the corresponding coordinate of $\mathbf{\Lambda}^\top \boldsymbol{\beta}$, and a BLUE if $\text{Cov}(\mathbf{BY} + \mathbf{d}) - \text{Cov}(\mathbf{AY} + \mathbf{c})$ is nonnegative definite for every LUE $\mathbf{BY} + \mathbf{d}$ (Definition 1.7.1).

The words “for every $\boldsymbol{\beta}$ ” in (b) carry all the weight. An estimator that is unbiased at one parameter value only, like the constant 0.5 when $\beta_1 = 0.5$, does not qualify. The first result (Proposition 7.1.1) turns unbiasedness into linear equations.

Proposition 7.1.1 — Unbiasedness as a linear equation

The linear estimator $c + \mathbf{a}^\top \mathbf{Y}$ is an LUE of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ iff $c = 0$ and $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$. Consequently $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ has an LUE iff $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$.

Proof. By Theorem 2.2.1, $E(c + \mathbf{a}^\top \mathbf{Y}) = c + \mathbf{a}^\top \mathbf{X}\boldsymbol{\beta}$. This equals $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ for all $\boldsymbol{\beta}$ iff the affine function $\boldsymbol{\beta} \mapsto c + (\mathbf{X}^\top \mathbf{a} - \boldsymbol{\lambda})^\top \boldsymbol{\beta}$ vanishes identically. Putting $\boldsymbol{\beta} = \mathbf{0}$ gives $c = 0$. Putting $\boldsymbol{\beta}$ equal to each coordinate vector in turn then gives $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$. The converse is immediate. An LUE therefore exists iff the equation $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$ has a solution, that is, iff $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$. ■

So the constant c is always zero, and from now on we drop it. A linear function $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ with $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ is called **estimable**. For these functions the data carry linear information. For the others no linear combination of the observations is centred on the target. Chapter 8 takes this as the definition of estimability (Definition 8.2.1) and develops its many equivalent forms. For now we need only Proposition 7.1.1 and one comparison. Theorem 6.4.4 of Chapter 6 found the same condition $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ as the answer to a different question: when is $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ the same for every least squares estimate $\hat{\boldsymbol{\beta}}$? The functions that least squares estimates unambiguously are exactly the functions that have linear unbiased estimators at all. When $\mathbf{X}$ has full column rank, $\mathcal{C}(\mathbf{X}^\top) = \mathbb{R}^p$, so every linear function of $\boldsymbol{\beta}$ is estimable.

7.1.3 The geometry of the unbiased class

Fix an estimable $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ and write $\boldsymbol{\lambda} = \mathbf{X}^\top \boldsymbol{\rho}$ for some $\boldsymbol{\rho} \in \mathbb{R}^n$. The coefficient vectors of its LUEs form the set

$$\mathcal{A}_\lambda = \{\mathbf{a} \in \mathbb{R}^n : \mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}\}.$$

This set is the key object of the chapter’s first half.

Proposition 7.1.2 — The set of unbiased coefficient vectors

Let $\boldsymbol{\lambda} = \mathbf{X}^\top \boldsymbol{\rho}$. Then:

- (a) $\mathcal{A}_\lambda = \mathbf{M}\boldsymbol{\rho} + \mathcal{C}(\mathbf{X})^\perp$, an affine subspace of $\mathbb{R}^n$ of dimension $n - r$, parallel to $\mathcal{C}(\mathbf{X})^\perp$;
- (b) every $\mathbf{a} \in \mathcal{A}_\lambda$ has the same projection $\mathbf{M}\mathbf{a} = \mathbf{M}\boldsymbol{\rho}$, and $\mathbf{M}\boldsymbol{\rho}$ is the only element of $\mathcal{A}_\lambda$ that lies in $\mathcal{C}(\mathbf{X})$;
- (c) $\mathbf{M}\boldsymbol{\rho} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^- \boldsymbol{\lambda}$ for every generalized inverse, and $(\mathbf{M}\boldsymbol{\rho})^\top \mathbf{Y} = \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ for every

least squares estimate $\hat{\beta}$.

Proof.

- (a) By [Proposition 7.1.1](#), $\mathbf{a} \in \mathcal{A}_\lambda$ means $\mathbf{X}^\top \mathbf{a} = \lambda$. Now $\mathbf{X}^\top \mathbf{a} = \mathbf{X}^\top \boldsymbol{\rho}$ iff $\mathbf{X}^\top (\mathbf{a} - \boldsymbol{\rho}) = \mathbf{0}$, iff $\mathbf{a} - \boldsymbol{\rho} \in \mathcal{N}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X})^\perp$ (Lemma 6.2.1). So $\mathcal{A}_\lambda = \boldsymbol{\rho} + \mathcal{C}(\mathbf{X})^\perp$. Since $\boldsymbol{\rho} - \mathbf{M}\boldsymbol{\rho} = (\mathbf{I} - \mathbf{M})\boldsymbol{\rho} \in \mathcal{C}(\mathbf{X})^\perp$, the same set is $\mathbf{M}\boldsymbol{\rho} + \mathcal{C}(\mathbf{X})^\perp$. Its dimension is $\dim \mathcal{C}(\mathbf{X})^\perp = n - r$.
- (b) If $\mathbf{a} = \mathbf{M}\boldsymbol{\rho} + \mathbf{v}$ with $\mathbf{v} \in \mathcal{C}(\mathbf{X})^\perp$, then $\mathbf{M}\mathbf{a} = \mathbf{M}\boldsymbol{\rho} + \mathbf{M}\mathbf{v} = \mathbf{M}\boldsymbol{\rho}$. If moreover $\mathbf{a} \in \mathcal{C}(\mathbf{X})$, then $\mathbf{a} = \mathbf{M}\mathbf{a} = \mathbf{M}\boldsymbol{\rho}$.
- (c) By Theorem 6.3.7, $\mathbf{M}\boldsymbol{\rho} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\rho} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \lambda$. By Theorem 6.4.4, $\lambda^\top \hat{\beta} = \boldsymbol{\rho}^\top \mathbf{M}\mathbf{Y} = (\mathbf{M}\boldsymbol{\rho})^\top \mathbf{Y}$.

■

Part (c) says that least squares is itself one of the linear unbiased estimators. Its coefficient vector is $\mathbf{a}_* = \mathbf{M}\boldsymbol{\rho}$, the one member of the class that lies in the model space. It does not depend on which $\boldsymbol{\rho}$ was used to represent λ , because every such $\boldsymbol{\rho}$ has the same projection. That $\mathbf{a}_*^\top \mathbf{Y}$ is unbiased can also be seen directly: $E(\mathbf{a}_*^\top \mathbf{Y}) = \boldsymbol{\rho}^\top \mathbf{M}\mathbf{X}\beta = \boldsymbol{\rho}^\top \mathbf{X}\beta = \lambda^\top \beta$.

The variance of any linear estimator follows from Theorem 2.2.1:

$$\text{Var}(\mathbf{a}^\top \mathbf{Y}) = \mathbf{a}^\top (\sigma^2 \mathbf{I}) \mathbf{a} = \sigma^2 \|\mathbf{a}\|^2. \quad (7.1.2)$$

Under the second-moment assumptions, then, the variance of a linear estimator is σ^2 times the squared length of its coefficient vector. Finding the BLUE means finding the shortest vector in the affine set $\mathcal{A}_\lambda$.

Idea. The linear unbiased estimators of $\lambda^\top \beta$ correspond to the points of a flat $\mathcal{A}_\lambda = \mathbf{a}_* + \mathcal{C}(\mathbf{X})^\perp$ in $\mathbb{R}^n$. The variance of each is σ^2 times its squared distance from the origin. The question “which LUE is best?” is a nearest-point problem, the same kind of problem that least squares solves in [Chapter 6](#), now posed in the space of coefficient vectors.

A second description of the class is useful too. Every $\mathbf{a} \in \mathcal{A}_\lambda$ is $\mathbf{a}_* + (\mathbf{I} - \mathbf{M})\mathbf{w}$ for some $\mathbf{w}$, so every LUE has the form

$$\mathbf{a}^\top \mathbf{Y} = \lambda^\top \hat{\beta} + \mathbf{w}^\top (\mathbf{I} - \mathbf{M})\mathbf{Y} = \lambda^\top \hat{\beta} + \mathbf{w}^\top \hat{\epsilon}. \quad (7.1.3)$$

An LUE is the least squares estimate plus a linear function of the residuals. The linear functions of the residuals are exactly the linear unbiased estimators of zero ([Exercise \(B1\)](#)). Adding one to an estimator keeps it unbiased and can only add noise. [Section 7.2](#) makes “can only add noise” precise.

7.1.4 Three estimators of a slope

Example. (Three linear unbiased estimators of a slope)

A straight line $E(Y_i) = \beta_0 + \beta_1 x_i$ is fitted to $n = 12$ observations at the regressor values

$x = 1, 2, 2.5, 3, 4, 5, 5.5, 7, 8, 9, 10.5, 12$, for which $S_{xx} = \sum_i (x_i - \bar{x})^2 = 137.229$. Here $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$ has full rank and the target is $\boldsymbol{\lambda} = (0, 1)^\top$. Three natural estimators of the slope are linear in $\mathbf{Y}$.

- (i) *Least squares*, $\hat{\beta}_1 = \sum_i (x_i - \bar{x})Y_i / S_{xx}$ (Theorem 5.2.1), with $a_i = (x_i - \bar{x}) / S_{xx}$.
- (ii) *End points*, the slope of the line through the first and last observations, $(Y_{12} - Y_1) / (x_{12} - x_1)$.
- (iii) *Group means*, which splits the data into the six smallest and six largest x values and takes $(\bar{Y}_{\text{high}} - \bar{Y}_{\text{low}}) / (\bar{x}_{\text{high}} - \bar{x}_{\text{low}})$.

Each coefficient vector satisfies $\mathbf{1}^\top \mathbf{a} = 0$ and $\mathbf{x}^\top \mathbf{a} = 1$, which is the condition $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$ of Proposition 7.1.1. So all three are unbiased, whatever the distribution of the errors. By Eq. 7.1.2 their variances are $\sigma^2 \|\mathbf{a}\|^2$:

$$0.00729 \sigma^2, \quad 0.01653 \sigma^2, \quad 0.01008 \sigma^2.$$

Least squares has the smallest, and the other two have efficiencies 0.441 and 0.723 relative to it. The listing also confirms what Proposition 7.1.2(b) predicts: the three coefficient vectors differ, but their projections onto $\mathcal{C}(\mathbf{X})$ all equal the least squares vector.

```
import numpy as np

x = np.array([1.0, 2.0, 2.5, 3.0, 4.0, 5.0, 5.5, 7.0, 8.0, 9.0, 10.5, 12.0])
n = len(x)
X = np.column_stack([np.ones(n), x])
lam = np.array([0.0, 1.0]) # target: the slope beta_1
M = X @ np.linalg.solve(X.T @ X, X.T) # projection onto C(X)

# coefficient vectors a of three linear estimators a^T y of the slope
a_ls = X @ np.linalg.solve(X.T @ X, lam) # least squares: a = X (X^T X)^{-1}
    lambda
a_end = np.zeros(n)
a_end[[0, -1]] = [-1.0, 1.0]
a_end /= x[-1] - x[0] # slope through the two end points
low, high = np.arange(n) < n // 2, np.arange(n) >= n // 2
a_grp = (high / high.sum() - low / low.sum()) / (x[high].mean() - x[low].mean()) #
    group means

for name, a in [("least squares", a_ls), ("end points", a_end), ("group means",
    a_grp)]:
    print(f"{name:14s} X^T a = {X.T @ a}    Var/sigma^2 = {a @ a:.5f}    Ma = a_ls: {
    np.allclose(M @ a, a_ls)}")
```

[Run this code online](#)

The end-point estimator throws away ten of the twelve observations, so its poor showing is no surprise. The group-means estimator uses every observation, and it is still clearly worse than least squares. It weights all observations in a half equally, while least squares weights each observation in proportion to its distance from $\bar{x}$. Estimators of this grouping type were proposed by Wald (1940) for regressions in which x itself is measured with error. Chapter 24 returns to that problem.

7.1.5 Why these restrictions?

Linearity and unbiasedness are restrictions of convenience as well as of principle, and it is worth being clear about what each one does.

Unbiasedness removes estimators that are good only near one parameter value. It is not sacred. [Section 7.2](#) shows biased estimators with smaller mean squared error, and [Section 7.6](#) derives a whole family of them from prior information. Unbiasedness can also be too demanding: a nonestimable $\lambda^\top \beta$ has no linear unbiased estimator, and under normality no unbiased estimator of any kind ([Exercise \(C1\)](#)).

Linearity is what makes the second-moment assumptions enough. The variance of a nonlinear estimator, such as a median or a trimmed mean, depends on the whole error distribution. Its variance can't be compared with that of least squares until that distribution is specified. For linear estimators [Eq. 7.1.2](#) needs only σ^2 . The restriction therefore has a cost: optimality among linear estimators says nothing about nonlinear ones. [Section 7.2](#) gives an error distribution under which a simple nonlinear estimator is far better than least squares. [Section 7.4](#) shows that under normality this cannot happen.

7.1.6 Exercises

7.1.6.1 A. Check your understanding

Exercise A1.

In the location model $Y_i = \mu + \varepsilon_i$ ($\mathbf{X} = \mathbf{1}$), show that the LUEs of μ are the weighted averages $\sum_i a_i Y_i$ with $\sum_i a_i = 1$. Identify $\mathcal{A}_\lambda$ and $\mathbf{a}_*$ and find the LUE of smallest variance directly.

Exercise A2.

In the one-way layout $E(Y_{ij}) = \mu + \alpha_i$ with three groups and an intercept (Example (Four answers, one fit)), show that α_1 has no linear unbiased estimator. Write down two different LUEs of $\alpha_1 - \alpha_2$, and verify that their coefficient vectors have the same projection onto $\mathcal{C}(\mathbf{X})$.

7.1.6.2 B. Practice

Exercise B1.

A linear estimator $\mathbf{a}^\top \mathbf{Y}$ is an **unbiased estimator of zero** if $E(\mathbf{a}^\top \mathbf{Y}) = 0$ for every β . Show that this holds iff $\mathbf{a} \in \mathcal{C}(\mathbf{X})^\perp$, iff $\mathbf{a}^\top \mathbf{Y}$ is a linear function of the residual vector $\hat{\varepsilon}$. Show that every unbiased estimator of zero is uncorrelated with $\lambda^\top \hat{\beta}$ for every estimable $\lambda^\top \beta$.

Solution. By [Proposition 7.1.1](#) with $\lambda = \mathbf{0}$, $\mathbf{a}^\top \mathbf{Y}$ is unbiased for zero iff $\mathbf{X}^\top \mathbf{a} = \mathbf{0}$, iff $\mathbf{a} \in \mathcal{C}(\mathbf{X})^\perp$. If so, $\mathbf{a} = (\mathbf{I} - \mathbf{M})\mathbf{a}$ and $\mathbf{a}^\top \mathbf{Y} = \mathbf{a}^\top (\mathbf{I} - \mathbf{M})\mathbf{Y} = \mathbf{a}^\top \hat{\varepsilon}$. Conversely, $\mathbf{w}^\top \hat{\varepsilon} = ((\mathbf{I} - \mathbf{M})\mathbf{w})^\top \mathbf{Y}$ has coefficient vector in $\mathcal{C}(\mathbf{X})^\perp$. For the last claim, write $\lambda^\top \hat{\beta} = \rho^\top \mathbf{M}\mathbf{Y}$. By Theorem 2.2.1, $\text{Cov}(\rho^\top \mathbf{M}\mathbf{Y}, \mathbf{a}^\top \mathbf{Y}) = \sigma^2 \rho^\top \mathbf{M}\mathbf{a} = 0$, because

$$M\mathbf{a} = \mathbf{0}.$$

Exercise B2.

Let $\mathbf{A}$ be $p \times q$ with $\mathcal{C}(\mathbf{A}) \subseteq \mathcal{C}(\mathbf{X}^\top)$. Show that $\mathbf{A}\mathbf{Y}$ is an LUE of $\mathbf{A}^\top\boldsymbol{\beta}$ iff $\mathbf{A}\mathbf{X} = \mathbf{A}^\top$, and that the set of such $\mathbf{A}$ is $\{\mathbf{A}^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top + \mathbf{W}(\mathbf{I} - \mathbf{M}) : \mathbf{W} \in \mathbb{R}^{q \times n}\}$.

7.1.6.3 C. Going deeper

Exercise C1.

Assume $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ and $\boldsymbol{\lambda} \notin \mathcal{C}(\mathbf{X}^\top)$. Show that no statistic $\delta(\mathbf{Y})$, linear or not, satisfies $E \delta(\mathbf{Y}) = \boldsymbol{\lambda}^\top\boldsymbol{\beta}$ for all $\boldsymbol{\beta}$. *Hint:* the distribution of $\mathbf{Y}$ depends on $\boldsymbol{\beta}$ only through $\mathbf{X}\boldsymbol{\beta}$.

Solution. Since $\boldsymbol{\lambda} \notin \mathcal{C}(\mathbf{X}^\top) = \mathcal{N}(\mathbf{X})^\perp$, there is $\mathbf{v} \in \mathcal{N}(\mathbf{X})$ with $\boldsymbol{\lambda}^\top\mathbf{v} \neq 0$. The parameters $\boldsymbol{\beta}$ and $\boldsymbol{\beta} + \mathbf{v}$ give the same mean $\mathbf{X}\boldsymbol{\beta}$, hence the same distribution of $\mathbf{Y}$, hence the same value of $E \delta(\mathbf{Y})$. An unbiased δ would need $\boldsymbol{\lambda}^\top\boldsymbol{\beta} = \boldsymbol{\lambda}^\top(\boldsymbol{\beta} + \mathbf{v})$, which is false. The argument uses only that the law of $\mathbf{Y}$ is a function of $\mathbf{X}\boldsymbol{\beta}$ and σ^2 . [Chapter 8](#) calls this *nonidentifiability* (Definition 8.1.1).

Exercise C2.

For the group-means estimator of [Example \(Three linear unbiased estimators of a slope\)](#) with general $x_1 \leq \dots \leq x_n$ ($n = 2m$), find its variance in terms of $\bar{x}_{\text{high}} - \bar{x}_{\text{low}}$. Show that its efficiency relative to least squares is $(\bar{x}_{\text{high}} - \bar{x}_{\text{low}})^2 n / (4S_{xx})$, and that this is 1 iff the x values take only two distinct values, with half of the observations at each.